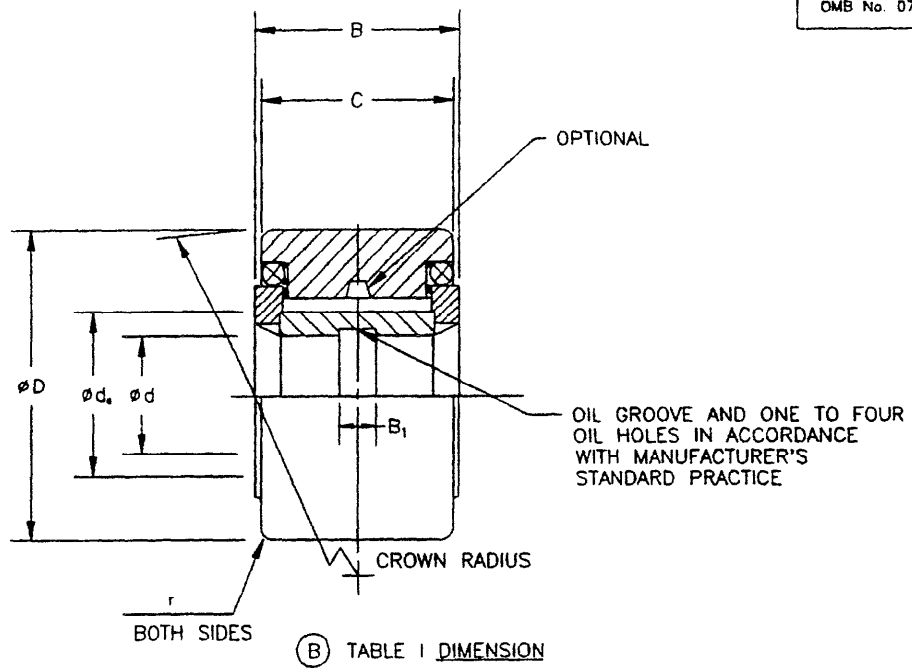
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THIS SPECIFICATION IS APPROVED FOR USE BY ALL DEPARTMENTS AND AGENCIES OF THE DEPARTMENT OF DEFENSE.

Form Approved
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(B) TABLE I DIMENSION

Dash No.	ød Bore	øD Outer Ring Outside Diameter	C Outer Ring Width	B Over-all Width	ød₀ Clamping Diameter Min.	r Rad Min. 1/	B1 Lubrication Groove Width	Crown Radius Ref.	Total Radial Clearance Max.	Capacity As a Track Roller lb.	Track Capacity 40 HRC lb.	Limit Load Rating lb.	Mass (Approx) lbs.
-103	.1900	.7500	.280	0.312	0.438	0.022	0.094	10.000	0.0018	900	290	1200	0.03
-104	.2500	.8750	.345	0.375	0.518	0.022	0.125	10.000	0.0018	1430	575	1910	0.05
-106	.3750	1.0625	.455	0.500	0.672	0.022	0.188	10.000	0.0018	2700	1000	3600	0.10
-108	.5000	1.3125	.580	0.625	0.844	0.032	0.188	12.500	0.0018	4300	1785	5780	0.18
-110	.6250	1.5000	.705	0.750	0.953	0.032	0.250	17.500	0.0018	6400	2600	8500	0.28
-112	.7500	1.7500	.950	1.000	1.109	0.032	0.250	25.000	0.0018	18700	4050	14200	0.52
-114	.8750	2.0000	1.075	1.125	1.219	0.032	0.250	27.500	0.0018	14400	5350	19300	0.75
-120	1.2500	2.5000	1.200	1.250	1.825	0.032	0.375	30.000	0.0018	18900	7100	25300	1.16
-124	1.5000	3.0000	1.440	1.500	2.000	0.032	0.375	60.000	0.0018	28400	10900	37900	2.36
-128	1.7500	3.4375	1.440	1.500	2.281	0.032	0.375	60.000	0.0018	33000	12400	44100	2.71
-132	2.0000	3.8750	1.440	1.500	2.562	0.032	0.375	60.000	0.0018	36700	14000	48900	3.42
-136	2.2500	4.3125	1.440	1.500	2.859	0.032	0.375	60.000	0.0020	41200	15600	55000	4.23
-140	2.2000	4.7500	1.440	1.500	3.109	0.032	0.375	60.000	0.0020	44900	17200	59900	5.14
-144	2.7500	5.0000	1.440	1.500	3.344	0.032	0.375	60.000	0.0020	46800	18100	64800	5.49

1/ MINIMUM CORNER RADIUS FOR OUTSIDE DIAMETER.

(B) TABLE II. TOLERANCE VALUES.

DIMENSIONS IN INCHES

ød Basic Bore		Allowable Deviation From d of Single Mean Dia., d_{mp}		Allowable Deviation From Over-all Width B		Allowable Deviation From Lubrication Groove Width B_1		ød Basic Bore		Allowable Deviation From D of Single Dia., d_{mp}		Allowable Deviation From Outer Ring Width C	
Over	Incl	High	Low	High	Low	High	Low	Over	Incl	High	Low	High	Low
0.1250	2.7500	0	-0.0007	0	-0.010	0	-0.062	0.6875	5.0000	0	-0.0007	0	-0.015

(B) DENOTES CHANGES

INCH-POUND

PREPARING ACTIVITY: NAVY-AS CUSTODIANS: ARMY-AT NAVY-AS AIR FORCE-99 REVIEW: ARMY-AT NAVY-AS AF-11,99 DLA-IS USER: PROJECT NUMBER 3110-0849	MILITARY SPECIFICATION SHEET TITLE BEARING, ROLLER, NEEDLE, SINGLE ROW, HEAVY DUTY, TRACK ROLLER, SEALED, TYPE V ANTIFRICTION, INCH	SPECIFICATION SHEET NUMBER MS21438 05 DEC 94 REV B SUPERSEDING MS21438A 31 MAR 81 AMSC- N/A FSC 3110
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REQUIREMENTS:

1. MATERIAL:

   a. STEEL, MIL-S-7493, MIL-S-8690, ASTM NO. A304, A322, A331, QQ-S-700, FED-STD-66, AISI/SAE STEEL NO. 50100, 52100, STEEL AMS 6640.

   b. SEALS AND BACKING RINGS - ACETAL RESIN PER L-P-392 OR NYLON PER L-P-410, TYPE 6/6, WEAR RESISTANT GRADE, OR POLYESTER ELASTOMER. CONSTRUCTION OPTIONAL.

2. FINISH OR PLATING:

   a. OUTER RING - OUTER RING CHROME PLATED PER QQ-C-320, CLASS 2. O.D. AND O.D. CORNERS .0005-.0010 INCH THICKNESS, FACES MINIMUM .0003 IN THICKNESS.

   (B) b. INNER RING - ALL EXTERNAL SURFACES, EXCEPT THE INNER RING BORE, ZINC-NICKEL IN ACCORDANCE WITH AMS 2417, TYPE 2, OR CADMIUM PLATED IN ACCORDANCE WITH QQ-P-416, TYPE II, CLASS 2, WITH A THICKNESS OF .0003 TO .0006 INCHES.

   (B) c. ENDPLATES - ALL EXPOSED SURFACES ZINC-NICKEL IN ACCORDANCE WITH AMS 2417, TYPE 2, OR CADMIUM PLATED IN ACCORDANCE QQ-P-416, TYPE II, CLASS 2, WITH A THICKNESS OF .0003 INCHES TO .0006 INCHES.

3. LUBRICANT: MIL-G-81322 OR MIL-G-23827. ALL BEARINGS SHALL BE PREPACKED WITH GREASE CONFORMING TO MIL-G-81322 UNLESS OTHERWISE SPECIFIED. IF MIL-G-23827 IS REQUIRED, ADD THE SUFFIX G TO THE MS PART NUMBER.

4. MARKING: THE MARKING SHALL CONSIST OF THE MS PART NUMBER AND THE MANUFACTURER'S IDENTIFICATION IN ACCORDANCE WITH MIL-STD-130.

5. PART NUMBER: THE MS PART NUMBER SHALL CONSIST OF THE MS NUMBER PLUS THE DASH NUMBER.

   EXAMPLES: MS21438-103, MS21438-103G.

(B) 6. PACKAGING: BEARINGS SHALL BE INDIVIDUALLY PACKAGED TO THE REQUIREMENTS OF MIL-P-197. PACKAGE MARKED WITH MANUFACTURER'S NAME OR TRADEMARK, AND DATE OF LUBRICATION BY MONTH AND YEAR.

7. DIMENSIONS: ALL DIMENSIONS ARE AFTER PLATING AND ARE IN INCHES.

8. LOAD RATING: THE LIMIT LOAD RATING LISTED CAN BE DEFINED AS THE MAXIMUM RADIAL LOAD WHICH CAN BE APPLIED TO A BEARING WITHOUT IMPAIRING THE SUBSEQUENT FUNCTIONING OF THE BEARING. THE ULTIMATE OR STATIC FRACTURE LOAD RATING IS NOT LESS THAN 1.5 TIMES THE LIMIT LOAD RATING. THE LOAD RATING AS A TRACK ROLLER IS THE LOAD THE BEARING WILL CARRY AS A TRACK ROLLER FOR AN L-10 LIFE OF 20,000 REVOLUTIONS.

9. TEMPERATURE: BEARING OPERATING TEMPERATURE RANGE -65°F TO 250°F.

10. REMOVE ALL BURRS AND SHARP EDGES.

NOTES:

1. MS21438 SUPERSEDES MS24465.

2. DESIGN FEATURES: FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER DOCUMENTS REFERENCED HEREIN.

3. REFERENCED DOCUMENTS: REFERENCED DOCUMENTS ARE THE ISSUES IN EFFECT ON DATE OF INVITATIONS FOR BID.

PREPARING ACTIVITY: NAVY-AS CUSTODIANS: ARMY- AT NAVY-AS AIR FORCE- REVIEW: ARMY-AT NAVY-AS AF-11,99 DLA-IS USER: PROJECT NUMBER: 3110-0849	MILITARY SPECIFICATION SHEET TITLE BEARING, ROLLER, NEEDLE, SINGLE ROW, HEAVY DUTY, TRACK ROLLER, SEALED, TYPE V ANTIFRITION, INCH	SPECIFICATION SHEET NUMBER MS21438 05 DEC 94 REV B SUPERSEDING MS21438A 31 MAR 81 AMSC- N/A FSC 3110
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